

- 20** Two waves of equal frequency and amplitude are travelling in opposite directions along a stretched string. When they meet, they form a stationary wave with three nodes and two antinodes. The frequency of both waves is doubled and a new stationary wave is formed.

How many antinodes are there in the new stationary wave?

A 1

B 3

C 4

D 5

Ans: C

Standing wave with three nodes and two antinodes is as shown:



When frequency is doubled, the wavelength is halved. Standing wave is as shown:



Hence getting 4 antinodes.